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RESEARCH

Quality Evaluation of Integrated Distance Course for Special Education Based on Group Decision-Making Algorithm

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Abstract
Due to the significant differences among normal and disabled students, the classical evaluation method leads to a lack of consensus and balance in the course quality evaluation, resulting to lower quality and accuracy. To solve this problem, this paper proposes a group decision-making algorithm for the quality evaluation of integrated distance courses. Firstly, the proposed algorithm uses the K-means algorithm to cluster and collect the learning behavior data of students into predefined K clusters based on feature similarity. Secondly, the proposed algorithm constructs evaluation indicators of course quality from multiple perspectives under the support of multiple evaluation principles. Finally, all personal preferences of each decision-maker are brought together into one collective preference, which is used to rank the decision-making algorithms. In this paper, students are regarded as decision-makers, the integrated courses are regarded as decision-making options, the learning behaviors and preferences of students are regarded as attribute decision variables. Thus, the quality evaluation of integrated distance courses can be regarded as a group decision-making process. Experimental results show that the proposed method performs well in both consistency ratio and comprehensive score. The consistency ratio of our method always remains above 0.9, and the TOPSIS comprehensive score reaches a maximum of 0.937.

Keywords Group decision-making · Integrated distance course · Quality evaluation · Indicator system · Cluster · Collective preference

1 Introduction

Integrated distance education is a part of higher education model that combines traditional face-to-face teaching with modern distance education technology. This model makes use of the Internet, multimedia, virtual reality and other technologies to provide students with a flexible approach of learning, enabling them to be educated without the limitations of time and space. Today, with the social demand for diversified education and the pursuit of educational equity, the integration of special and general education has become an important direction of educational reform [1, 2]. However, the quality evaluation methods for special education in integrated education, especially the related distance education, have not yet been sufficiently focused on and researched.

The course quality evaluation can be applied to different groups of students, including special and general education. Such an evaluation method considers the requirements and characteristics of different kinds of students and ensures the fairness and effectiveness of the evaluation. The research on the evaluation methods of the quality of integrated education has important theoretical and practical significance [3–5].

Firstly, through the study of different course evaluation methods, it enables educational institutions to assess the teaching effect and student learning situation of the integrated courses accurately and provides a scientific basis for teaching; Second, the establishment of scientific and effective evaluation methods promotes the integration and interaction between special and general education, which promotes the overall development of education [6]. In

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addition, studying the course evaluation methods of the integration of special and general education promotes the rational allocation of educational resources, improves the quality of education and teaching, and contributes to the cultivation of inclusive and innovative talents [7]. Therefore, it has important practical significance and development prospects.

The current research on course evaluation methods is popular. Reference [8] proposes a course quality evaluation method based on the random forest algorithm. This method analyses the level of assisted teaching during college and completes the evaluation of the quality of multimedia-assisted teaching in these courses through the specification of the random forest algorithm and the optimization of the indicators of the evaluation of teaching quality. The random forest algorithm tends to overfitting when handling complex datasets, i.e., the model is so complex that it not only learns the patterns in the data but also learns the noise, leading to a decrease in the prediction performance. Reference [9] proposes a quality evaluation method for online courses based on FCE-AHP. Based on real cases of online courses, this method divides the factors influencing the quality of online courses into 4 first-level indicators, which are subdivided into 14 s-level indicators. The weight vectors of the evaluation indicators were determined based on the opinions of the experts of the "Academic Affairs Committee", and the fuzzy evaluation members were calculated based on the results of the questionnaire survey of 2021 students. However, the weight vectors in the method were determined by the experts of the Academic Affairs Committee, and this operation may introduce subjectivity. Experts' opinions may be influenced by personal experience, bias, or knowledge limitations, resulting in less than objective weight assignments. So, Reference [10] proposes an online course quality evaluation method based on the Consciousness Normal Cloud. This method analyses the cloud digital features of the Consciousness Normal Cloud and derives the distribution of the cloud digital features based on the weights of the course teaching evaluation, achieving a comprehensive evaluation of the online course quality. Cloud digital features are the core concepts in the cloud model, but they are not as intuitive as traditional statistical indicators, which easily leads to difficulties for educational decision-makers and course designers to accurately understand the meaning of the evaluation results. Reference [11] proposes a MOOC quality evaluation method based on the Grounded theory, which constructs an evaluation framework for the quality of the catechism and determines the overall goal of course teaching. Based on the selected MOOC quality evaluation indicators, the AHP method is used to determine the weight of each evaluation indicator and the fuzzy algorithm is used to evaluate the comprehensive evaluation indicators at each level. The quality evaluation results of English MOOC are derived based on Grounded theory. Grounded theory is a qualitative research method that relies on the researcher's subjective interpretation and analysis. When constructing the MOOC quality evaluation framework, the researcher's subjective judgment may affect the construction of the evaluation framework and the final evaluation results.

In order to solve the overfitting or strong subjectivity in existing evaluation methods, a group decision-making algorithm based evaluation system is introduced to construct a more objective, accurate, and easy to understand curriculum quality evaluation framework, which supports education decision-makers and curriculum designers to make more effective teaching improvement and resource allocation decisions. Through group decision-making, the influence of the subjective bias of a single decision-maker is reduced. Mutual discussion and weighing among group members balance personal bias and improve the objectivity and fairness of evaluation. Moreover, the group decision-making algorithm can be adjusted and optimized according to different evaluation needs and situations. This flexibility makes the evaluation process adaptable to the changing educational environment and course characteristics, ensuring the effectiveness of the evaluation method.

Detailed technical route of the quality evaluation in this paper is as follows:

(1) The K-means algorithm is used to cluster and analyze the learning behavior data of students in integrated courses. The algorithm divides the data into K predetermined clusters based on their feature similarity. Through the clustering process, the learning patterns and behavioral characteristics of different student groups are identified, providing a data foundation for subsequent course quality evaluation.
(2) Guided by the principle of multiple evaluations, a comprehensive and detailed curriculum quality evaluation index system is constructed from multiple perspec-

tives such as teaching content, student interaction, and resource availability. Ensure that the evaluation indicators cover all aspects of the course, in order to conduct a comprehensive and in-depth evaluation of the course quality.

(3) Considering students as decision-makers, integrating courses as decision-making plans, and students' learning behaviors and preferences as attribute decision variables, transforming the evaluation of teaching quality into a group decision-making process. By collecting individual preferences of each decision maker and integrating them into collective preferences, sorting and selecting the best among different decision-making methods can be achieved. Iterative optimization method is used to continuously adjust and optimize the decision-making processing until obtaining the optimal quality evaluation.

2 Course quality evaluation of integrated higher education

Integrated education is an educational philosophy that aims to enable students with different levels of ability and learning needs to learn, interact, and grow together. Integrated education focuses not only on general education students but also includes students with a variety of special needs and disabilities. This educational model ensures that each student can fully participate in the learning process and succeed by providing personalized support and resources. The objectives of integrated education are to promote mutual understanding, respect, and cooperation among students, to develop the social skills and humanistic qualities of all students, to break down barriers and discrimination in traditional education, and to achieve full inclusion in education. Students learning in higher integrated courses have a certain time sequence, so according to the sequence of courses, the learning behavior data extraction process is diagrammed in the form of a flowchart. The flowchart is shown in Fig. 1.

As can be observed in Fig. 1, the arrows are connected sequentially from left to right, revealing the sequence of course advancement. This indicates that the course advancement follows a certain sequence. Only when one course is completed will students move on to the next course and the associated data extraction process follow. The flowchart makes it possible to visually track the student's entire learning process from the beginning to the end of the course, with each step of the process visible briefly.

2.1 Filtering and classification of learning behavior data in integrated courses

When conducting big data attribute analysis of learning behaviors in the field of special education, the way of data collection, data visualization and interpretation, and data collection cycle and frequency are important factors affecting analysis accuracy. Adopting scientific and objective data collection, such as visual sensor technology and learning behavior monitoring systems, contributes to improving the accuracy of data collection. At the same time, reasonable data visualization and interpretation contribute to the understanding of students' learning behaviors and the development of personalized education programs. In addition, shorter collection cycles and higher collection frequency avoid missing and outdated data and

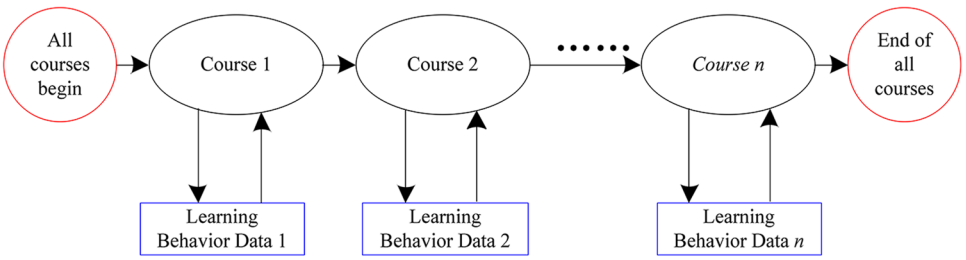


Fig. 1 Learning behavior data extraction process

improve the validity and accuracy of the data. The big data attributes of learning behaviors in the field of special education mainly involve the following aspects:

(1) Students' personal information: including gender, age, family background, physical condition, type of disability, etc.
(2) Academic performance data: including student grades, academic progress, course selection, etc.
(3) Learning behavior data: including study time, learning styles, learning strategies, homework completion, etc.
(4) Mental health data: including self-esteem, self-confidence, adaptability, etc.
(5) Extracurricular activities data: including participation in social activities, social skills, hobbies, etc.
(6) Educational resources utilization data: including library borrowing, laboratory usage, etc.

The data used is from higher university which has the integration for special and general education. The analyzed students' learning behaviors include online activities, homework submissions, discussion participation, interaction frequency, learning duration, and exam scores, etc., which are collected from online learning platforms, course management systems by user' tracking software. The data collection cycle covers 16 weeks, an entire semester, to ensure the continuity and integrity of the data. Data are collected once a week or two to observe the changes in students' learning behavior at continuous time points. To protect participants' privacy, all collected data are anonymized before analysis by removing identifiable personal information [12].

The collected learning behavior data contains a large amount of noise, such as accidental clicks, false operations, or system error records, which interferes with the analysis of students' real learning behavior. Also, the duplicate/highly similar data records during the long-term data collection processing increase the complexity of data processing and reduce analysis efficiency. Therefore, it is necessary to clean the learning behavior data.

This paper designs an improved filtering algorithm to narrow down the number of cluster candidates by determining whether a data cluster is a dynamic or static cluster and deciding whether it should be removed from the cluster candidate set. This improved filter can effectively remove noise and redundancy from the data by identifying and deleting dynamic or static clusters, which improves the overall quality of the data. Reduction of the cluster candidates' number can reduce the complexity of subsequent data analysis, accelerate data processing speed, and improve analysis efficiency; the deletion of inconsistent data clusters helps to improve the accuracy of data analysis, making the analysis results more similar to students' real learning behavior.

The algorithmic flow of the improved filtering algorithm is as following Algorithm 1.

Algorithm 1

Step 1.

For all candidate course evaluation-related datasets

Constructs a kd-tree;

Calculates $n.count$, $n.total$, $(x_{i,min}, x_{i,max}), i = 1,2,\ldots,d$ of each node in the tree;

Step 2.

Let $rt.SC = CC^0$;

// CC^0 is the center set of the initial cluster, $rt.SC$ are the candidate center sets of root notes

Step 3.

For each node n, exclude the root node:

Let $n.SC = pn.SC$; //$pn.SC$ are the candidate center sets of parent nodes of n

Determine the nearest neighbor center C_i of n's center;

Calculate $r_1^{'}$, $r_2^{'}$;

If n is a leaf node: Let C_i+= n;

Else

For each cluster center $C \in n.SC/\{C_i\}$:

Determine the vertex V;

If $d(V,C) > d(V,C_i)$:

Remove C from $n.SC$;

If $|n.SC| = 1$: Let C_i+= n;

Step 4.

Let $p = 1$, generate a new cluster center set CC^p; // p is Lloyd iterations' number

Step 5.

Determine the active cluster center CC_{active} from CC^p;

Let $rt.SC = CC^p$, $rt.SC_{active} = CC_{active}$, $C_i = 0$;

// $rt.SC_{active}$ is the center candidate set of active root node

Step 6.

For each node n exclude the root node

Let $n = pn$;

Determine the nearest neighbor cluster center from n, calculate $r_1^{'}$ and $r_2^{'}$;

If n is a leaf node: Let C_i+= n;

Elseif $C_i \neq C_i'$ and $d(\text{n. center}, C_i) \geq r_2'$

For each cluster center $C \in n.SC/\{C_i\}$

Determine the vertex V;

If $d(V,C) > d(V,C_i)$:

Remove C from $n.SC$;

If $|n.SC| = 1$: Let C_i+= n;

Else

If $C \in n.SC_{active}$ or $(C \in n.SC_{static}$ and $C \notin n.SC')$

Determine the vertex V;

If $d(V,C) > d(V,C_i)$:

Remove C from $n.SC$;

If $|n.SC| = 1$: Let C_i+= n;

Step 7.

Let $p = p + 1$;

Produce a new cluster center set CC^p

While the cluster center changes: Goto Step 5.

End Algorithm 1.

At each stage of the iteration, Algorithm 1 determines the closest cluster center C_i to n's center and checks whether each member in $n.SC/\{C_i\}$ needs to be clipped out. Step 6 mainly determines the nearest neighbor cluster center C_i of n and checks whether each member in $n.SC_{active}$ needs to be clipped out. It means that, at each stage of iteration, Step 6 reduces the calculation complexity by about $\sum |n.SC| / \sum |n.SC_{active}|$ times. Here, $n.SC$ and $n.SC_{active}$ are obtained by clipping some members from the cluster center set and the active cluster center, respectively. When $|n.SC|$ and $|n.SC_{active}|$ are proportional to $|n.SC|$ and $|n.SC_{active}|$ respectively, that is, $(\sum |n.SC| / \sum |n.SC_{active}|) \approx (|CC|/|CC_{active}|)$, we complete the data cleaning.

Because the students of integrated courses have big differences, the K-means algorithm is used to classify the learning behavior data, which is an unsupervised learning algorithm. So, K-means algorithm is suitable for analyzing and classifying the learning behavior data of the students because it doesn't need pre-labeled training [13]. Moreover, the K-means algorithm is simple and easy to implement, computationally efficient, and quickly processes large amounts of data and produces better clustering results. Therefore, when exploring and evaluating the course quality of special education in higher integrated education, the K-means algorithm is conducive to effectively analyzing the learning behavior data of students with large differences.

The K-means algorithm requires a dataset with many data samples and specific values for the number of clusters [14]. Next, use randomly selected samples as the initial cluster centers, and calculate the distance between the rest of the samples and the initial center to measure which sample is more similar to the initial center. Then place the sample in the cluster class where the centers are located. We continue the calculation based on the centers of the changing cluster classes and stop until the error is reduced to a minimum. Clustering calculation of learning behavior data of higher integrated education course using K-means algorithm is started given a sample set of K:

$$\begin{cases} K = \langle C_x \middle| C_x = \left(C_{x1}, C_{x2}, \ldots, C_{xj}\right) \rangle \\ x = 1,2,\ldots,n \end{cases} \tag{1}$$

where n denotes the sample capacity, C denotes the set attributes of the learning behavior data sample, and x and j denote that the $x-th$ data in the set K has j attributes.

After selecting the learning behavior data sample, the cluster class center is determined:

$$\begin{cases} B = \langle I_v \middle| I_v = \left(I_{v1}, I_{v2}, \ldots, I_{vd}\right) \rangle \\ v = 1,2,\ldots,m \end{cases} \tag{2}$$

where B denotes the set of all cluster class centers in learning behavior data, I_v denotes the set of center attributes, v and d denote that the center I_v has d different attributes, and there are m cluster classes that need to be calculated.

When both the set of learning behavior data and the cluster class centers are determined, the set of similarity distances between the two are calculated:

$$\begin{cases} X(C_x, I_v) = \sqrt{\sum_d (C_x - I_v)^2} \\ x = 1,2,\ldots,n; v = 1,2,\ldots,m \end{cases} \tag{3}$$

where $T(C_x, I_v)$ denotes the measured value of similarity distance between the learning behavior data and the cluster class center.

Table 1 Indicator system for quality evaluation of integrated course

Level 1 indicators	Introduction of indicators
Course Objectives and Content	To evaluate the scientific and rationale of the course design, including whether the objectives of the course are clear, and the content are matching with the needs of the field of special education.
Teaching Methods and Strategies	To evaluate the diversity and effectiveness of teaching methods and strategies, including teachers' teaching skills, use of teaching aids, and interactive teaching.
Student Learning Performance	To evaluate the effectiveness of students' knowledge acquisition, skills enhancement, cognitive development, and potential development in the course.
Course Evaluation and Adjustment	To evaluate the scientific and accuracy of course assessment methods, and the timely feedback and adjustment of the assessment results.
Teaching Resources Support	To assess the adequacy and effectiveness of teaching resources, including the availability of teaching resources such as libraries, laboratories, and special equipment.
Faculty Development	To evaluate the level of educational background, professional knowledge, and teaching ability of teachers, as well as the support and mechanism for their continuing professional development.
Student Participation and Satisfaction [19]	To evaluate the motivation and satisfaction of students in participating in the course, including classroom participation and extra-curricular activities participation.
Social Recognition and Effectiveness	To evaluate the extent to which the course facilitates students' future entry into the workplace, social participation, and social adaptability.

Based on the K-means algorithm, it quickly and efficiently classifies the learning behavior data of students. Based on the similarity merit to determine which cluster class a learning behavior belongs to, and minimize the classification error, provide an effective data base for the subsequent course quality evaluation.

2.2 Indicator system for courses quality evaluation

After completing the data collection of learning behavior in courses, the quality evaluation indicator system is constructed. Firstly, set the principles of higher integrated education course quality evaluation. The specifics are as follows:

(1) Individualization

Special education involves students with different types and degrees of disabilities, who have differences in learning ability, communication style, cognitive level, and so on. Therefore, the evaluation system should develop personalized evaluation criteria and methods based on the personal differences of the students. This may involve a comprehensive evaluation based on the type and degree of disability, understanding the student's needs and potential, and providing him/her with appropriate support and services.

(2) Comprehensiveness

Evaluation of special education courses needs to take into consideration all aspects of the course. This includes evaluating the reasonableness and clarity of the course objectives, e.g. whether they are in line with the needs in the field of special education; evaluating the design of the course content, e.g. whether it adequately meets the learning needs of the students; and evaluating the diversity and effectiveness of the teaching methodology, e.g. whether teaching strategies suitable for different students have been adopted.

(3) Operability

Evaluation indicators should be operable, i.e. they can be collected and measured objectively and are easy to evaluate implementation and improvement measures. This means that the evaluation indicators should be clear, measurable, and able to quantitatively or qualitatively reflect the quality and effectiveness of the special education courses.

(4) Sustainability

Evaluation of courses in special education should be an ongoing process that continues throughout the course. The evaluation process should include data collection, analysis of results and feedback, and provide support for continuous improvement of the courses. This facilitates early identification of problem-solving and sustained improvement in the quality of teaching and learning.

(5) Participation

Evaluation process should encourage the participation of teachers, students, parents, and other relevant stakeholders. Their comments and suggestions will provide valuable information to help evaluators fully understand and assess the quality and effectiveness of the special education courses [15]. Through cooperation and communication with these stakeholders, the objectivity and comprehensiveness of the evaluation are enhanced to ensure their consensus on the course.

(6) Transparency

Evaluation criteria and methods should be open and transparent, and all relevant parties should learn about the evaluation criteria and process. This ensures the fairness and credibility of the evaluation and enhances the participation of the evaluation. Transparency also improves teachers' and students' understanding and acceptance of evaluation and promotes their active participation and feedback [16].

The indicator system of course quality evaluation is an important tool for measuring the quality and effectiveness of courses in special education. This study constructs the indicator system of course quality evaluation referring to the existing literatures. Li et al. [17] constructed the indicator system from aspects of students' performance, sense of interest, sensibility, etc. Xu et al. [18] constructed the indicator system from the aspects of teaching course design, teaching method, teaching status, and learning effect.

Based on existing research and in-depth consideration of the course features, the indicator system is constructed by combining multiple dimensions such as learning objectives and standards, outcomes, process and methods, course evaluation and adjustment, learning resource support, faculty strength and development, student participation and satisfaction, social acceptance and effect. All these dimensions are used to evaluate whether the course design is scientific and reasonable, the teaching methods are diverse and effective, the student learning outcomes are significant, the course evaluation is timely and accurate, the teaching resources are sufficient and effective, the teaching staff is strong and continuously developing, the student's participation is active and satisfactory, the course has a positive impact on the students' future adaptability. Such a comprehensive evaluation provides a scientific basis for quality enhancement and continual improvement of courses. Detailed indicators are shown in Table 1.

In the evaluation indicator system, a comprehensive framework has been constructed, aiming to evaluate the effectiveness and quality of courses from multiple

dimensions. It provides a structured evaluation framework and uniform standards to the group decision-making algorithm, ensuring the scientific consistency of the evaluation process. Additionally, it facilitates the integration of data from different evaluators and enhances the objectivity and fairness of the following decision-making. Through this indicator system, the group decision-making algorithm conducts multidimensional in-depth analysis, optimizes the decision-making process, and supports the continuous improvement of course quality.

2.3 Course quality evaluation based on group decision-making algorithms

With the evaluation indicator system, the group decision-making algorithm is constructed to evaluate the quality of these courses, effectively enhancing the scientific and effective evaluation quality. In the evaluation, different subjects and consensus mechanisms have a profound impact on the effectiveness; diverse perspectives and expertise enhance the evaluation participation and representativeness, and ensure the alignment of courses with industry requirement; consensus mechanism improves the quality of decision-making, enhances the acceptance of evaluation results, encourages continuous improvement, and reduces personal bias by promoting communication and negotiation among subjects. These indicators together ensure the comprehensiveness, fairness, and practicality of the evaluation results, thereby promoting the continuous improvement of the courses quality for integrated distance education.

Group decision-making algorithm brings together the opinions and experiences of multiple evaluators, avoiding the subjectivity and bias of a single evaluator, and facilitating the comprehensive consideration of different people's opinions on course quality. The group decision-making algorithm promotes the democratic participation of teachers, students, parents, and other relevant people, increasing the fairness and credibility of the evaluation. Moreover, this algorithm promotes knowledge sharing and exchange, incorporating the expertise and experience of all parties to provide more accurate and comprehensive results for the evaluation. Most importantly, the group decision-making algorithm produces more accurate evaluation results by integrating the opinions and decisions of all parties, providing scientific support for the improvement and development of higher integrated courses.

In group decision-making theory, group decision-making is the process of bringing together the individual preferences of each decision-maker into collective preferences, so that the decision-making group ranks each decision-making algorithm as optimal. If students are regarded as decision-makers, higher integrated courses as decision-making solutions, and students' learning behaviors and preferences as attributed decision-making variables, then according to the description of the problem, the teaching quality evaluation of higher integrated courses is regarded as a group decision-making process.

In the process of higher integrated education, different learning behavior data of students are acquired with different scales, and to avoid the course quality evaluation being affected by different scales, learning behavior data need to be normalized [20, 21]. Let $X_{i,j}^1, X_{i,j}^2, \ldots, X_{i,j}^h$ denotes the observation data of students' learning behavior. Then the standardized processing formula for student learning behavior data is expressed as:

$$Y_{i,j}^k = \frac{X_{i,j}^k - min_{\forall k}\{X_{i,j}^k\}}{max_{\forall k}\{X_{i,j}^k\} - min_{\forall k}\{X_{i,j}^k\}} \tag{4}$$

Where:

$$min_{\forall k}\{X_{i,j}^k\} = min\{X_{i,j}^1, X_{i,j}^2, \ldots, X_{i,j}^h\} \tag{5}$$

$$min_{\forall k}\{X_{i,j}^k\} = max\{X_{i,j}^1, X_{i,j}^2, \ldots, X_{i,j}^h\} \tag{6}$$

The more students are involved in the teaching and learning process of higher integrated courses, the better the learning performance achieved. Similarly, different learning behaviors of students have different effects on the course quality evaluation. Therefore, we set the weights of students so that students with better performance of learning behaviors, are weighted higher importance of evaluation results in group decision-making.

Based on the standardized learning behavior data $Y_{i,j}^k$, the mean value of the student's $j-th$ learning behavior is calculated separately:

$$Ts_j^k = \frac{1}{n}\sum_{i=1}^{n} Y_{i,j}^k \tag{7}$$

At the same time, each student's participation in the $j-th$ learning behavior is ranked in descending order and recorded as:

$$\sigma_j = \{Ts_j^{1'}, Ts_j^{2'}, \ldots, Ts_j^{2'}\} \tag{8}$$

Where $Ts\alpha_j \geq Ts\beta_j (1 \leq \alpha < \beta \leq h)$.

The weights of the students were calculated through formula (9):

$$\omega_j^k = Q\left(\frac{\sum_{\alpha=1}^{k} Ts_j^{\alpha}}{\sum_{\beta=1}^{h} Ts_j^{\beta}}\right) - Q\left(\frac{\sum_{\alpha=1}^{k-1} Ts_j^{\alpha}}{\sum_{\beta=1}^{h} Ts_j^{\beta}}\right) \tag{9}$$

Where $\sum_{\alpha=1}^{k} Ts_j^{\alpha}$ denotes the sun of the first k values in σ_j ,which is $\sum_{\alpha=1}^{k} Ts_j^{\alpha} = Ts_j^{1'}, Ts_j^{2'}, \ldots, Ts_j^{2'}$, $Q(X)$ is a monotonic

function made of weighted aggregates on a basic unit interval.

Based on the student's participation Ts_j^k, find its corresponding value $Ts_j^{k'}$ in σ_j, and then calculate student's weights ω_j^k according to the formula.

The weights of the students are normalized to let:

$$\sum_{k=1}^{h} \omega_j^k = 1 \tag{10}$$

Where h denotes the number of students.

It is worth noting that students may have the same value of participation, but only one value is taken in the rank σ_j, which aims to ensure that students with the same level of participation have the same weights.

Using the normalized raw data and the weights of the students, to construct the group decision matrix $R = (r_{ij})_{n\times m}$, with each element of the matrix R being:

$$r_{ij} = \sum_{k=1}^{h} Y_{i,j}^k \omega_j^k \tag{11}$$

Each student's quality evaluation of higher integrated courses needs to be considered comprehensively to see whether his evaluation results reach a coherent consensus with other students. Only when students reach a coherent consensus on the decision-making can the quality of higher integrated courses be evaluated objectively, fairly, and accurately [22]. The consensus index is calculated from three levels: students' learning behavior, educational courses, and students' decision-making.

(1) Consensus index of learning behaviors and preferences

The consensus index of the $j-th$ learning behavior of the $i-th$ teaching course of student s^k relative to the group is:

$$E_{ij}^k = 1 - \frac{\left| r_{ij} - Y_{i,j}^k \right|}{2} \tag{12}$$

(2) Consensus index of educational courses

The consensus index of the $i-th$ educational course of student s^k relative to the group is

$$A_{ij}^k = \frac{1}{m} \sum_{j=1}^{m} E_{ij}^k \tag{13}$$

(3) Consensus index of student decision-making

The consensus index of student s^k on the decision matrix relative to the groups is:

$$Z^k = \frac{1}{n} \sum_{i=1}^{m} A_{ij}^k \tag{14}$$

Table 2 Evaluation results of higher integrated courses quality

Evaluation results	Evaluation levels
$\bar{r}_i \geq 0.9$	Excellent
$0.75 \leq \bar{r}_i < 0.9$	Good
$0.6 \leq \bar{r}_i < 0.75$	Fair
$\bar{r}_i < 0.6$	Substandard

The larger value of student s^k consensus index $Z^k (0 \leq Z^k \leq 1)$, the greater the possibility that student s^k will reach a coherent consensus with other students. Letting the formula index critical value κ, if the consensus index of the students is all greater than or equal to the consensus index critical value, indicates that these students have reached a coherent consensus; otherwise, it is necessary to adjust the consensus index.

For students who have not reached a coherent consensus, their consensus index needs to be adjusted so that we can evaluate course quality.

Firstly, students who did not reach a coherent consensus need to be identified. If the consensus index Z^k of student s^k less than κ, it means that the student has not reached a coherent consensus, denoted as:

$$U = \left\{ s^k \middle\{ Z^k < \kappa \right\} \tag{15}$$

Secondly, it is identified in which educational course the student has not reached a coherent consensus. For a student in U, if $A_{ij}^k < \kappa$, it means that he/she has not reached coherent consensus in the $i-th$ educational course, denoted as:

$$P = \left\{ \left(s^k, i \right) \middle| s^k \varepsilon U \wedge A_i^k < \kappa \right\} \tag{16}$$

Finally, it is identified on which learning behavior or preference the student has not reached a coherent consensus. For each tuple (s^k, i) in P, it is also necessary to determine that the student has not reached a coherent consensus on the level of learning behavior concerning the preference, denoted as:

$$F = \left\{ \left(s^k, i, j \right) \middle| \left(s^k, i \right) \in P \wedge E_{ij}^k < \kappa \right\} \tag{17}$$

With the tuple (s^k, i, j) in F, it is possible to accurately obtain which student has not reached a coherent consensus in which learning session of which educational course. During the first round of adjustment, it is possible to find out which student has problems and deficiencies in which learning session of which educational course. At this point, the information needs to be fed back to teachers and

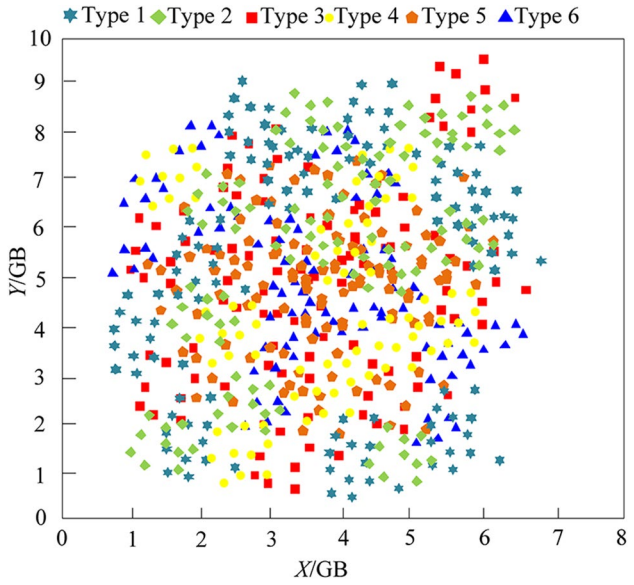


Fig. 2 Diagram of distribution of learning behavior data

students. Based on the feedback information, teachers and students analyze the reasons for students' learning difficulties in higher integrated courses, help students improve their learning methods, and adjust the course content to improve the quality of students' learning.

The elements (s^k, i, j) in F are adjusted to the student's learning behavior data through Eq. (18):

$$(Y_{i,j}^k)^* = \delta \cdot r_{ij} + (1-\delta) Y_{i,j}^k \tag{18}$$

In the formula, δ denotes the consensus adjustment factor.

After adjusting the students' consensus index, the students' consensus index is recalculated until all students form a coherent consensus.

When all students reach a coherent consensus, the course quality emulation of the higher integrated education can be carried out. The effective evaluation indicators set is $f = (f_1, f_2, \ldots, f_m)$, and its corresponding weights are $w = (w_1, w_2, \ldots, w_m)$, and $\sum_{j=1}^{m} w_m = 1$. Then the course quality evaluation of all students is expressed as:

$$\bar{r}_i = \sum_{j=1}^{m} r_{ij} \times w_{ij} \tag{19}$$

According to the results of the course quality evaluation obtained by formula (19), it is possible to evaluate the quality of higher integrated courses hierarchically, and the evaluation results correspond to the evaluation levels shown in Table 2.

3 Experimental validation

To validate the effectiveness of the proposed quality courses evaluation method based on a group decision-making algorithm for higher integrated education, comparative experiments are carried out.

3.1 Sample data

The sample data used in this experiment came from an institution that includes special education in higher integrated education, and the learning behavior data came from a semester of a 16-week higher integrated education course with 100 students. The detailed types of samples are as follows:

Type 1: Student engagement data: student learning behavior is evaluated by observing student engagement in the classroom. The number of times students ask questions, the accuracy and depth of their answers, and the frequency of their participation in group discussions or teamwork are recorded.
Type 2: Learning performance data: Learning performance is evaluated through exams, assignments, projects, and so on. These data reflect the level of performance of students in terms of the knowledge, skills, and abilities they have learned in the course.
Type 3: Feedback data: teachers regularly collect feedback from students on the content and methods of teaching. It is obtained through verbal feedback, questionnaires, or grading.
Type 4: Learning process data: learning time data is obtained by students completing classroom activity sheets or recording the amount of time students spend learning in the classroom. This data includes the time students spend studying independently within the classroom as well as the time they spend discussing and collaborating with their classmates. In addition, observing how often students use the learning resources provided in the classroom (e.g., textbooks, courseware, lab equipment, etc.) is data. Recording the amount of time students spend completing and delivering assignments, enables this research to get the situation of how well students are completing their assignments as well.
Type 5: Learning Interaction Data: it focuses on the interaction and communication between students, including the number of posts and replies in online discussion forums, the frequency of private message

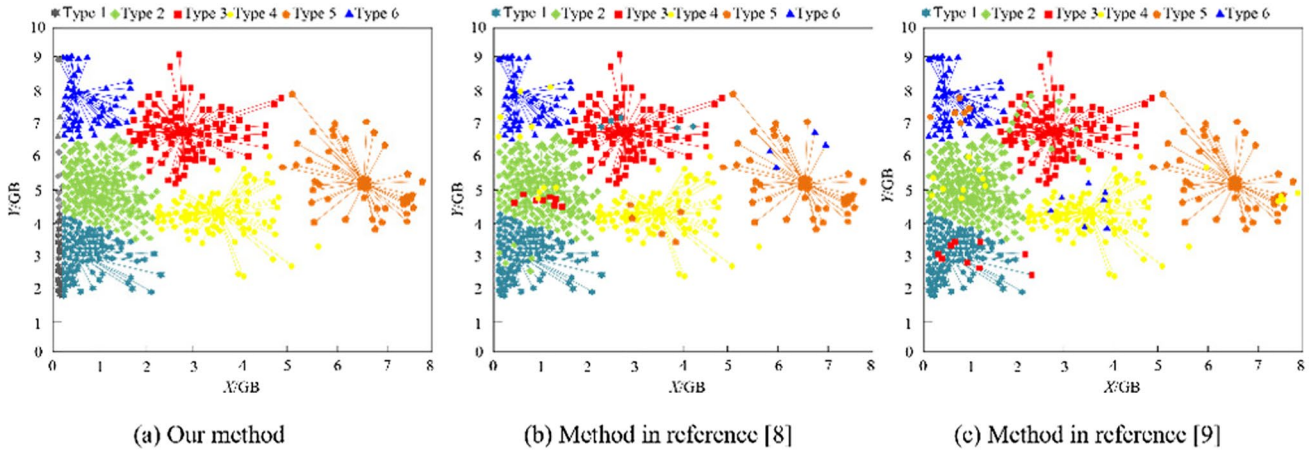


Fig. 3 Clustering effect of learning behavior data by different methods

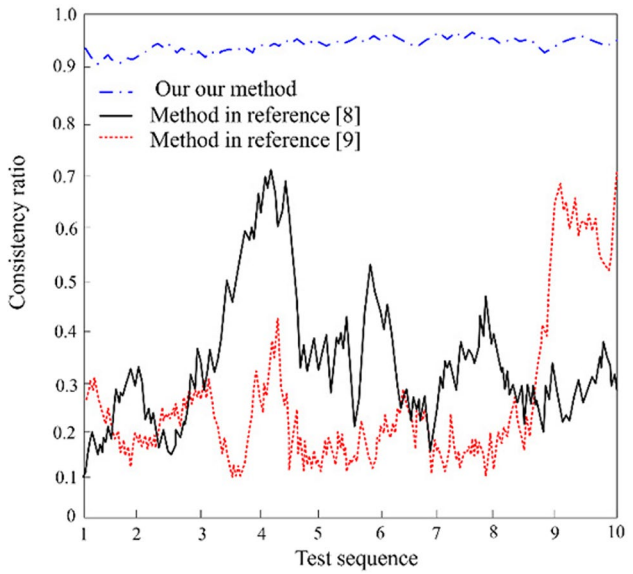


Fig. 4 Consistency ratio results

exchanges, and real-time interactions in virtual learning environments. These reflect the collaboration, social skills, and social learning behaviors among students.

Type 6: Technology usage data: the technology usage habits have a significant impact on students' learning experience and effectiveness, including the frequency of logging into the learning platform, the types of devices used (such as computers, tablets, and mobile phones), the stability of network connections, and familiarity with the platform functions. By analyzing these data, we can understand students' adaptability and utilization efficiency of technical tools, thereby optimizing technical support and services.

To observe the distribution of learning behavior data more intuitively, this experiment plots data points in a two-dimensional coordinate space, where X and Y indicate the size of the storage space for learning behavior data. The diagram of the distribution of learning behavior data under the two-dimensional coordinate space X and Y is shown in Fig. 2.

As can be seen in Fig. 2, the six types of data points are evenly distributed throughout the two-dimensional coordinate space, with no obvious aggregation or discrete trends. Different types of data points are mixed, and there is no obvious classification boundary. Therefore, to effectively use learning behavior data, it is necessary to accurately classify different types of learning behavior data into clusters.

In order to ensure the reproducibility of the experiment, the K-means algorithm clusters the dataset with following parameters: the number of clusters was set to 6; the initial clustering centers are randomly selected to reduce the possible differences in clustering results; Euclidean distance is the measurement; the maximum number of iterations is set to 100; the stopping condition is that the change in the cluster center is less than 0.0001, random seed is set to 42.

3.2 Experimental solution and indicators

This experiment takes learning behavior data clustering effect, consistency ratio, and TOPSIS overall score as the test indicators. We compare and verify our method with the method of reference [8] and the method of reference [9].

Learning behavior data clustering effect: assess whether the clustering algorithm classifies learners with similar

Table 3 TOPSIS overall score

Test number	TOPSIS overall score		
	Our method	Method in reference [8]	Method in reference [9]
1	0.832	0.615	0.521
2	0.824	0.603	0.519
3	0.917	0.658	0.543
4	0.825	0.687	0.512
5	0.914	0.621	0.596
6	0.937	0.694	0.574
7	0.882	0.629	0.523
8	0.908	0.611	0.535
9	0.895	0.602	0.501
10	0.843	0.616	0.589

learning behaviors into the same class. Within classes, the differences between learners should be as small as possible.

Consistency ratio: the consistency ratio is an indicator derived by comparing the consistency factor to a random consistency indicator. It provides a quantitative way of evaluating the consistency of the comparisons provided by the decision maker in the judgment matrix.

TOPSIS Overall Score: The overall score derived from the TOPSIS (Technique for Order Preference by Similarity to Ideal Solution) allows for a comprehensive consideration of the strengths and weaknesses of group decision-making algorithms as an overall indicator for evaluating performance.

3.3 Experimental results analysis

3.3.1 Clustering effect of learning behavior data

Analyzing learning behavior data through clustering helps identify and understand the learning characteristics, tendencies, and habits of different student groups. This helps educators better understand student groups and provides references for personalized teaching and tutoring. Testing learning behavior data through clustering is used to evaluate the quality of higher integrated courses and to understand the learning and feedback of different groups. This helps to identify strengths and room for improvement in course design and enhance the quality of teaching and learning. The clustering effect of learning behavior data by different methods are shown in Fig. 3.

By analyzing Fig. 3, it is found that our method outperforms the methods in reference [8] and reference [9] in terms of the clustering effect of learning behavior data. Specifically, our method clusters learning behavior data more accurately, showing a better clustering effect. This indicates that our method has higher clustering accuracy and effectiveness in dealing with learning behavior data, which is superior to existing methods. The comprehensive graphical results not only confirm this method has excellent performance in learning behavior data clustering but also highlight its potential and advantages in the course quality evaluation of higher integrated education, which provides useful references and insights for further research and practice.

The K-means algorithm allocates data to predetermined K clusters through an iterative optimization process for the similarity between them. Because of the efficiency and simplicity, K-means can quickly and effectively group data when processing large-scale learning behavior data. In this article, precise feature selection and calculation make the data are most similar within each cluster, while most different between clusters. This precise feature selection and computation improve the accuracy of clustering, making the proposed method perform well in learning behavioral data clustering.

3.3.2 Consistency ratio

The consistency ratio test is used to evaluate the consistency of group decision-making algorithms in the course quality evaluation of higher integrated education. The consistency ratio test helps to determine whether the decision makers are logically consistent in evaluating to avoid subjective bias and inconsistency. The consistency ratio test measures the degree of consistency between factors or indicators in a group decision-making algorithm. When the consistency ratio is low, there may be inconsistent factors in the decision-making process that need to be further optimized and adjusted to improve the reliability of the decision-making. The results of the consistency ratio test for the above three methods are shown in Fig. 4.

According to the analysis results shown in Fig. 4, it is clearly that our method performs significantly better in terms of consistency ratios than the methods mentioned in reference [8] and reference [9]. Specifically, the consistency ratios of the methods in this paper are generally higher than 0.9, showing a high level of consistency. This indicates that our method performs well in the consistency of evaluation results, and has higher reliability and stability compared with other methods. The comprehensive data shown in the graph not only verifies the excellent performance of this paper's method in terms of consistency ratio but also highlights its significant advantages in the evaluation of the course quality of higher integrated education, which provides substantial support and guidance for further exploring the application and promotion of the method.

The proposed method adopts the evaluation based on group decision-making, which can integrate the opinions of multiple decision makers and gather individual preferences

into collective preferences through certain rules and procedures. This improves the consistency and rationality of decision-making, ensures the consistency of evaluation results, and makes the evaluation between different decision-makers more coordinated and unified.

3.3.3 TOPSIS comprehensive scoring

The TOPSIS makes a comprehensive evaluation of multiple evaluation indicators and gets a comprehensive score by calculating the weights and scores among the indicators. In the course quality evaluation of higher integrated education, several aspects of indicators may be involved, and TOPSIS helps to take these indicators into account in a comprehensive way and provide comprehensive evaluation results. The results of the TOPSIS overall scores of the three methods are shown in Table 3.

Analyzing the experimental data in Table 3, the advantage of our method over other methods in terms of TOPSIS overall score is visualized. Taking test number 3 as an example, the score of our method is 0.917, compared with 0.658 and 0.543 for the methods of reference [8] and reference [9], respectively. This indicates that our method achieves significantly higher overall scores under this test, which shows that it is more capable of evaluating and consistently performing well across multiple test samples.

This is because the proposed method adopts the principle of multiple evaluations to comprehensively consider course quality from multiple perspectives when constructing the evaluation indicators. This multi-view evaluation reduces the bias from single-view evaluation, improves the comprehensiveness and accuracy of the evaluation results, and thus achieves higher scores in the TOPSIS comprehensive score.

4 Conclusion

This paper proposed a course quality evaluation of higher integrated education based on the group decision-making algorithm, which demonstrates significant advantages in the consistency ratio test and the TOPSIS overall score test. The consistency test ensures that the evaluation is fair and objective, the TOPSIS test considers multiple indicators comprehensively. Therefore, this method has the advantages of accuracy, stability, and comprehensiveness in course quality evaluation, which provides a scientific decision-making basis for higher education administrators. It promotes the continuous improvement of higher education quality.

The novelty of the proposed method is as follows:

(1) Novel application of group decision-making
This article constructs a group decision-making process by using students as decision makers, integrated courses as decision plans, and students' learning as decision attributions. It is more democratic and objective, better reflecting the common opinions and preferences of student groups, thereby improving the consistency and accuracy of evaluation decisions.

(2) Effectiveness of K-means algorithm
Experimental results show that the K-means algorithm has high clustering accuracy and effectiveness in processing learning behavior data. Through cluster analysis, this method can effectively identify the learning different student groups, providing strong data support for personalized teaching and tutoring.

(3) Better TOPSIS and Consistency
Experimental results show that the proposed method performs well in TOPSIS overall scoring and consistency ratio. This demonstrates the role of the mult-view evaluation and preferences integration, which more accurately reflect various aspects of course quality and avoid subjective bias and inconsistency.

However, further research is still needed to improve the method in the future. We will explore empirical applications and comparisons with other evaluation methods, to promote the effectiveness and sustainable development of the method in practice.

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Data Availability No datasets were generated or analysed during the current study.

Declarations

Competing Interests The authors declare no competing interests.

References

1. Aziz EAAA (2021) Design architecture of an integrated student activities management system for higher education. Turkish J Comput Math Educ 12(5):1676–1683
2. Patra KC, Basantia TK (2021) Integrated programmes in education: development and current status. Higher Educ Future 8(2):180–196

3. Alowayr A, Al-Azawei A (2021) Predicting mobile learning acceptance: an integrated model and empirical study based on higher education students' perceptions. Australas J Educ Technol 37(3):38–55
4. He X, Zhou Y, Luo D et al (2024) Education resource recommendation based on graph neural network and multi-subject rating. Comput Eng 50(7):13–22
5. Megan CB, Kate J, Jenny W et al (2023) Evaluation of the Interprofessional Spiritual Care Education Curriculum in Australia: Online. Palliat Support Care 21(1):65–73
6. Maldonado M, Farouk S, Campbell K et al (2021) Development, implementation and evaluation of a Limited English proficiency Curriculum. J Sci Innov Med 4(2):30
7. Niemi H (2021) Education reforms for equity and quality: an analysis from an educational ecosystem perspective with reference to finnish educational transformations. Cent Educ Policy Stud J 11(2):13–35
8. Liu G, Zhuang H (2022) Evaluation model of multimedia-aided teaching effect of physical education course based on random forest algorithm. J Intell Syst 31(1):555–567
9. Tang B, Guo S, Yeboah M et al (2021) Quality evaluation of online courses during COVID-19 pandemic based on integrated FCE-AHP method. J Intell Fuzzy Syst 41(1):1487–1498
10. Wang M (2023) A teaching evaluation method of English online course based on conscious normal cloud model. Int J Contin Eng Educ Life-long Learn 33(1):1
11. Ren S, Yu B (2022) The evaluation method for English MOOC quality based on grounded theory. Int J Contin Eng Educ Life-long Learn 32(1):1
12. Li Y, Zhu J, Fu W (2022) Intelligent privacy Protection of End user in Long Distance Education. Mob Netw Appl 27(3):1162–1173
13. Rani LN, Defit S, Muhammad LJ (2021) Determination of student subjects in Higher Education using Hybrid Data Mining Method with the K-Means Algorithm and FP Growth. Int J Artif Intell Res 5(1):91–101
14. He G, Vialle S, Baboulin M (2021) Parallel and accurate k-means algorithm on CPU-GPU architectures for spectral clustering. Concurr Comput: Pract Exp 34(14):e6621
15. Ergin A (2021) Evaluation of the third grade life science course curriculum according to eisner's educational criticism model. J Int Educ Sci 26(8):1–18
16. Zulema NLH (2021) Accreditation and certification as key factors of quality in higher education in engineering schools. High Educ Res 6(1):11–19
17. Li K, Zhang B, Meng Y et al (2023) Research on teaching quality evaluation model of engineering economics based on SOM neural network. Mod Electron Techn 46(18):162–166
18. Xu Z, Wang J, Wang Y et al (2023) Research on quality evaluation of online courses for we media platforms based on deep learning. E-Educ Res 44(08):42–48
19. Liu S, Peng C, Srivastava G (2023) What influences computational thinking? A theoretical and empirical study based on the influence of learning engagement on computational thinking in higher education. Comput Appl Eng Educ 31(6):1690–1704
20. Wu T, Chen L, Du G et al (2023) CDNM: clustering-based data normalization method for automated vulnerability detection. Comput J 67(4):1538–1549
21. Huang Y, Peng C, Liu S (2024) Empirical research of classroom behavior based on online education: a systematic review. Mob Netw Appl Online First. https://doi.org/10.1007/s11036-023-02251-2
22. Verma R, Mittal A (2023) Multiple attribute group decision-making based on novel probabilistic ordered weighted cosine similarity operators with pythagorean fuzzy information. Granul Comput 8(1):111–129

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